

SERVER:-

DATA CENTER:-SERVER ROOMS

IT IS A COMPUTER PROGRAM WHICH IS DESIGNED TO SHARE THE RESOURCE TO MULTIPLE CLIENTS.

CLIENT-SERVER:-

WEB BROWSER->CLIENT

SERVER->COMPUTER PROGRAM

->HANDLE THE INCOMING REQUEST FROM THE CLIENT AND GIVE RESPONSE TO THE CLIENT.

->DATA WILL BE STORED OVER THE SERVER

HTTP->FOR DATA COMMUNICATION WE HAVE PROTOCOL WHICH WE WILL CALL HTTP PROTOCOL.

REQUEST:-

RESPONSE

APPLICATION:-IT IS A SOFTWARE OR COMPUTER PROGRAM WHICH IS USED TO PERFORM SOME SPECIFIC TASK.

1.DESKTOP APPLN:-

1.IT NEEDS INSTALLATION

2.IT WILL RUN IN THE GIVEN SYSTEM OR COMPUTER

3.IT WILL USE COMPUTER RESOURCES

4.WE DONT NEED INTERNET FOR DESKTOP APPLN

5.WE CAN ACCESS WITHIN SYSTEM

6.PERFORMANCE WILL BE FAST

7.WE CAN NOT ACCESS FROM ANOTHER DEVICE & WE CANT UPDATE EASILY THE APPLN

EX:-MS WORD,CALCULATOR,PHOTOSHOP

2.WEB APPLN:-

1.WE CAN ACCESS WEB APPLN WITHIN THE BROWSER

2.WE NEED INTERNET FOR ACCESSING WEB APPLN

3.IT WILL RUN OVER THE SERVER

4.WE DONT NEED ANY INSTALLATION FOR WEB APPLN

5.WE CAN ACCESS ANYWHERE

6.WE CAN UPDATE APPLN VERY EASILY AND FAST

THIS IS DONE BY SERVER ITSELF

7.WEB APPLN:-INSTAGRAM,TWITTER,PHONEPAY,GPAY,RAPIDO

->WEB APPLN HAS BEEN DEVELOPED WITH THE HELP OF SERVER SIDE PROGRAMMING LANG.

->SERVLET,SPRING,MVC,BOOT,JSP,JAVASCRIPT

->SERVLET:-SERVLET IS A SERVER SIDE TECHNOLOGY,WHICH IS USED TO CREATE WEB APPLN.

->(SERVLET BE PRESENT OVER THE SERVER,GENERATE DYNAMIC WEBPAGE)

->WEB PAGE:-IT INCLUDES IMAGES,VIDEO,AUDIO

->COLLECTION OF WEB PAGES IS CALLED AS WEB SITE.

->1.STATIC WEB PAGE,2.DYNAMIC WEB PAGE

2.SERVLET IS AN API WHICH WILL HAVE SO MANY BUILT IN CLASSES AND INTERFACES.

EX:-SERVLET INTERFACE,GENERIC SERVLET,HTTP SERVLET ETC.

3.IT WILL HAVE ALL THE CAPABILITIES OF SERVER.

4.SERVLET IS DEVELOPED WITH THE HELP OF JAVA.

ADVANTAGES:-

1.IT IS SECURE.

2.IT IS ROBUST.

3.PERFORMANCE IS FAST.

4.IT WILL SUPPORT MULTITHREADING AS WELL.

*CLIENT-SERVER ARCHITECTURE:-

1.CLIENT->BROWSER,CHROME,OPERA MINI,FIRE FOX

2.SERVER->WEB SERVER,APPLN SERVER

CLIENT---->SEND REQUEST---->SERVER

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 CLIENT <--- RESPONSE

SERVLET API:-

1.WE HAVE TO DOWNLOAD APPACHE TOMCAT

2.HOW TO CREATE SERVLET PROJECT

SERVLET API:-

IN SERVLET API WE HAVE BUILT IN CLASSES ALONG WITH THAT BUILT IN INTERFACES.

EXAMPLE:-

1.SERVLET INTERFACE

2.GENERIC SERVLET CLASS

3.HTTP SERVLET CLASS

4.REQUEST DISPATCHER

5.HTTP SERVLET REQUEST

6.HTTP SERVLET RESPONSE ETC.

->IT IS PRESENT IN JAVAX.SERVLET PACKAGE.

->SERVLET INTERFACE IS THE PARENT FOR ALL THE CLASSES AND INTERFACES IN SERVLET API.

->IT WILL PROVIDE FEW FUNCTIONALITY,WHICH EVERY SERVLET HAS TO INHERIT.

->SERVLET INTERFACE WILL HAVE 5 MAIN BUILT IN METHODS.

->IN WHICH 3 METHODS ARE LIFE CYCLE METHODS.

LIFE CYCLE:-

1.SERVLET CLASS WILL BE LOADED WITH THE HELP OF CLASSLOADER

2. CREATING INSTANCE OF SERVLET CLASS

3. invoke init() method by web container

4. service method by web container

5. destroy() method by web container

1. GENERIC SERVLET:- IT IS A SERVLET CLASS WHICH IS PRESENT IN SERVLET API.

-> IT IMPLEMENTS SERVLET INTERFACE.

-> IT IS PROTOCOL INDEPENDENT.

-> IT CAN HANDLE ANY TYPE OF REQUEST.

-> IT WILL IMPLEMENT ALL THE METHODS OF SERVLET INTERFACE.

-> LIKE init() method, service(), destroy() method and so on.

-> WE CAN CREATE SERVLET WITH THE HELP OF GENERIC SERVLET CLASS ALSO.

2. HTTP SERVLET:- IT IS A SERVLET CLASS WHICH IS PRESENT IN SERVLET API.

-> IT IS A CHILD CLASS FOR SERVLET INTERFACE.

-> IT IS PROTOCOL DEPENDENT.

-> IT WILL HANDLE ALL HTTP REQUEST AND RESPONSE.

-> IT WILL INHERIT ALL THE METHODS OF SERVLET INTERFACE.

-> IT WILL HAVE doGet(), doPost(), doHead(), doTrace() methods.

3. REQUEST DISPATCHER:- IT IS AN INTERFACE WHICH IS PRESENT IN SERVLET API.

-> REQUEST DISPATCHER INTERFACE PROVIDES THE FACILITY TO DISPATCH THE REQUEST TO ONE RESOURCE TO ANOTHER RESOURCE.

-> IT MAY BE HTML, SERVLET OR JSP.

-> IN THIS INTERFACE WE HAVE TWO METHODS:-

1. FORWARD() METHOD:- IT FORWARDS A REQUEST FROM SERVLET TO ANOTHER RESOURCE

(SERVLET, JSP OR HTML FILE).

2. INCLUDE() METHOD:- IT INCLUDES THE CONTENT OF RESOURCE IN THE RESPONSE.

4. SENDREDIRECT() METHOD:- IT IS A METHOD OF HTTP SERVLET RESPONSE INTERFACE

IT CAN BE USED TO REDIRECT RESPONSE TO ANOTHER RESOURCE, IT MAY BE SERVLET OR JSP PAGE.

5. SERVLET CONFIG:- SERVLET CONFIG IS AN INTERFACE WHICH IS PRESENT INSIDE SERVLET API.

AN OBJECT OF SERVLET CONFIG IS CREATED BY WEB CONTAINER FOR EACH SERVLET.

THIS OBJECT IS USED TO GET CONFIGURATION INFORMATION FROM WEB.XML FILE.

6. SERVLET CONTEXT:- IT IS AN INTERFACE, WHICH IS PRESENT INSIDE SERVLET API.

->AN OBJECT OF SERVLET CONTEXT IS USED TO GET CONFIGURATION INFORMATION FROM WEB.XML FILE.

7.HTTP SESSION:-SESSION TRACKING IS A WAY TO MAINTAIN A DATA OF AN USER.

->IT IS ALSO KNOWN AS SESSION MANAGEMENT.

->IN HTTP SESSION CONTAINER CREATES A SESSION ID FOR EACH USER.

THE CONTAINER USES THIS ID TO IDENTIFY THE PARTICULAR USER.

AN OBJECT OF HTTP SESSION IS USED TO TRACK THE SESSION OF A USER.

->FOR SESSION TRACKING WE HAVE SEVERAL OPTIONS:-

1.USER AUTHENTICATION

2.URL REWRITING

3.HTML HIDDEN FIELD

4.COOKIES

1.GET:-

2.POST